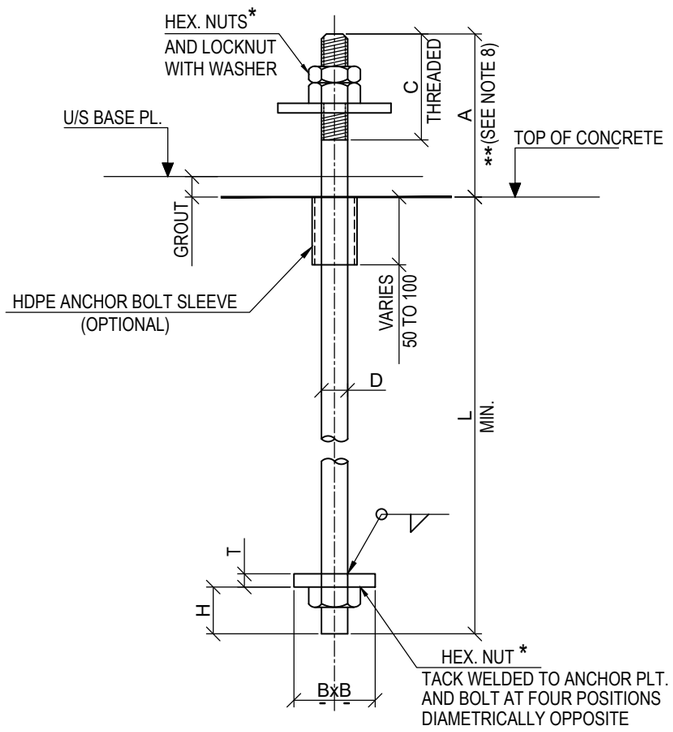
DRAWING NUMBER MD-216-1000-EG-CV-DED-1003 REV D01



TYP. DETAIL OF BOLT

SCHEDULE OF BOLTS

NOMINAL BOLT SIZE	D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM)				
			L MIN.	A	C	T	H
M16	16	50x50	350	150	100	10	30
M20	20	60x60	450	150	100	10	30
M24	24	75x75	600	175	125	12	30
M24	25	75x75	600	175	125	12	30
M30	30	100x100	800	200	150	16	45
M32	32	100x100	800	200	150	16	45
M32, M33	33	100x100	800	200	150	16	45
M36	36	110x110	1000	225	175	20	45
M39	39	120x120	1000	225	175	20	50
M40	40	120x120	1000	225	175	20	50
M45	45	150x150	1100	250	200	25	55
M50, M52	50, 52	150x150	1250	275	200	25	60
M56	56	165x165	1400	275	200	25	65
M63	63	180x180	1600	325	250	28	75
M80	80	240x240	1850	375	300	40	90
M100	100	300x300	2250	400	325	50	110

LEGEND:-
- TYP. = TYPICAL
- HEX. = HEXAGONAL
- U/S = UNDER SIDE
- MIN. = MINIMUM

NOTES:-
1. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE FOR RCC WORK CONSIDERED TO BE M35 UNLESS OTHERWISE NOTED
3. ALL BOLTS SHALL CONFORM TO ASTM F1554 GRADE 36.
*4. ALL NUTS SHALL CONFORM TO ASTM A563, GRADE A HEX/ HEAVY HEX.
5. WASHERS SHALL BE IN GENERAL SUPPLIED ALONG WITH STRUCTURE TO SUIT THE DIA OF HOLE PROVIDED IN STRUCTURAL BASE PLATE.
6. WASHERS SHALL BE ASTM F436 OR EQUIVALENT IF SUPPLIED WITH ANCHOR BOLT.
7. DIA OF HDPE SLEEVE SHALL BE MIN. 2.5xDIA OF BOLT. PIPE SLEEVE SHALL CONFORM TO ASTM D3350
**8. PROJECTION LENGTH FOR TANK ANCHOR BOLTS SHALL BE FOLLOWED AS PROVIDED IN RESPECTIVE TANK FOUNDATION G.A. DRAWINGS.
9. THE CONTENTS OF THIS DRAWING SHALL CONFORM TO PROVISION OF DOC. # MD-216-1000-EG-ST-CRT-1001.

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	15/03/2023	AS-BUILT	SYA	SJM	SOR	AG	
C04	26/03/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C04	JS	SJM	SOR	AG	
C03	11/01/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03	SYA	SJM	SOR	AG	
C02	04/06/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02	ARU	SJM	SOR	AG	
C01	07/03/2020	ISSUED FOR CONSTRUCTION	ARU	SJM	SOR	AG	
A01	30/12/2019	ISSUED FOR INFORMATION	ARU	SJM	AVG	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: O19045_900_C_D_24352_001 — REV 5
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZZ01 _CTB17 _10003 — DOCUMENT ID: OU100666798 — REV 5

DRAWN	ARU	18/12/2019
CHECKED	SJM	19/12/2019
ENGINEER	AVG	20/12/2019
APPROVED	AG	21/12/2019
EMPLOYER		
SCALE	1:100	A3

MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
ANCHOR BOLTS
STANDARD DETAILS

DRAWING NUMBER: MD-216-1000-EG-CV-DED-1003 — REV D01